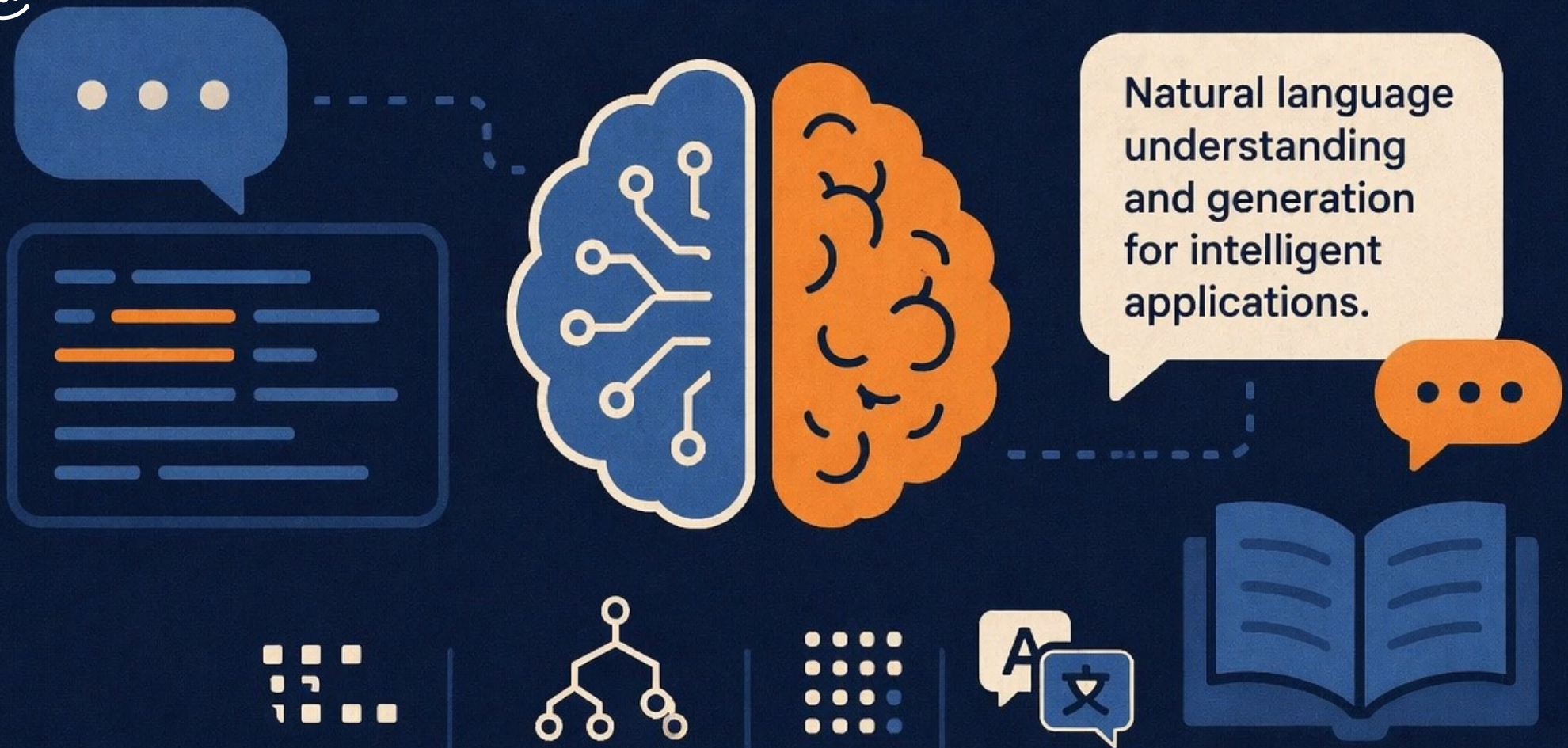
NATURAL LANGUAGE PROCESSING

class starts @ 9:05 pm



Lecture: LSTM

Long
Short
Term
Memory

#Agenda

1) What is LSTM?
2) Coding

RNN (problem)

1) Vanishing / Exploding gradient

chain of words is longer

2) Sequential data Input

Training N/w

Types of RNN

1) Many to One — Sentiment Analysis

2) Many to Many — Translation, sumization, chatbot

Inshorb

↳ News app

(Summarize the whole news 60 words)

i/p — Many to Many — o/p

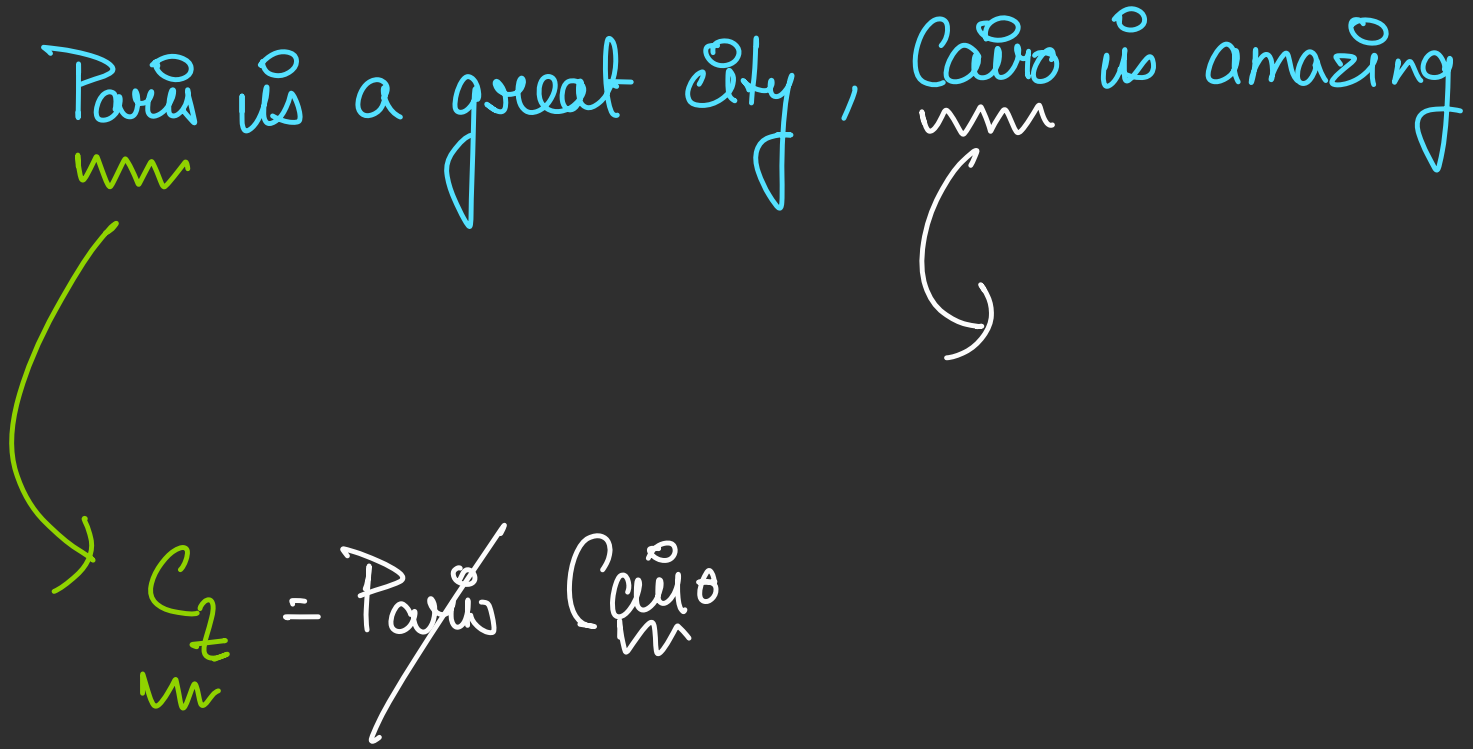
raw/whole news

↳ 60 word summary

"Encoder - decoder"

How

Short summary.

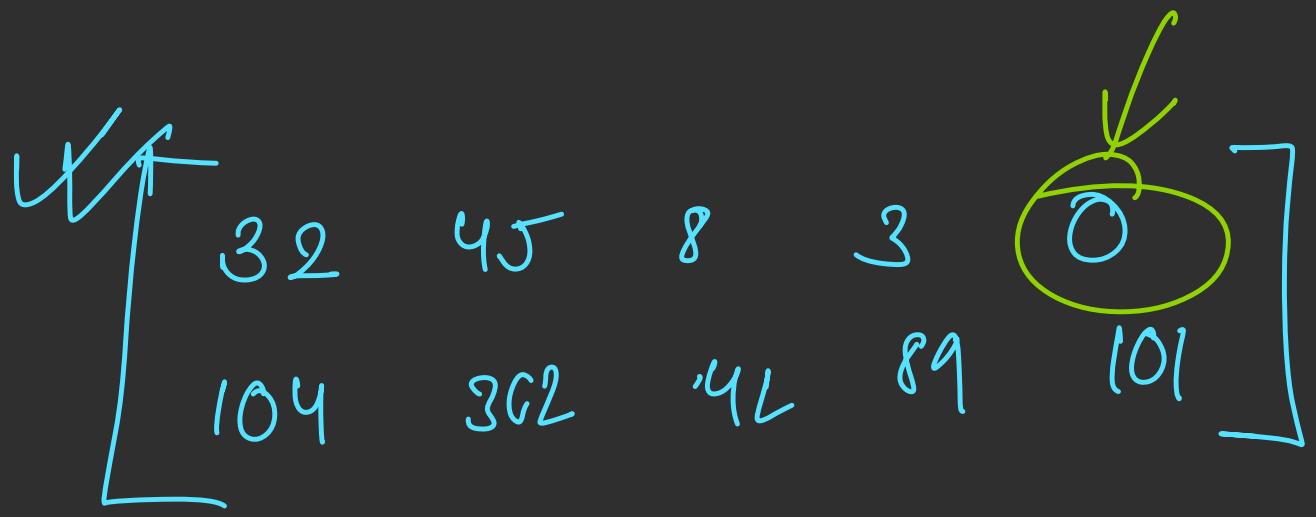

Paris is a great city, Cairo is amazing
C_t = Paris Cairo
Output → Paris

I have a cat → [32 45 8 3]

There is my father going → [104 302 42 89 101]

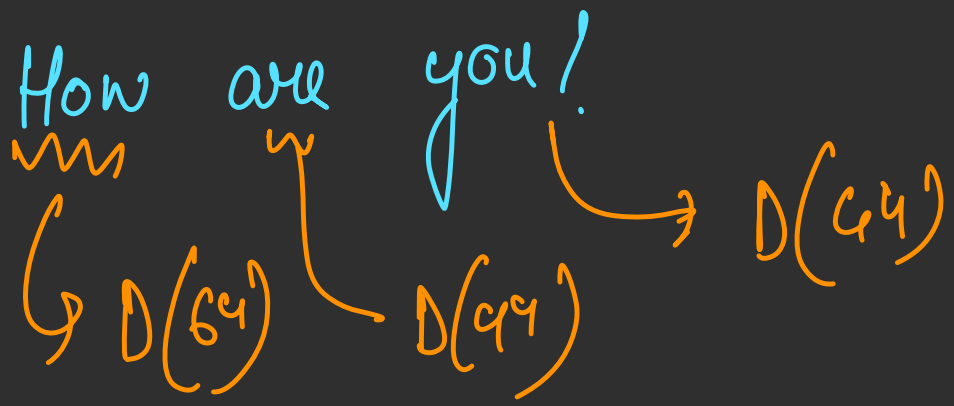
Batch prow

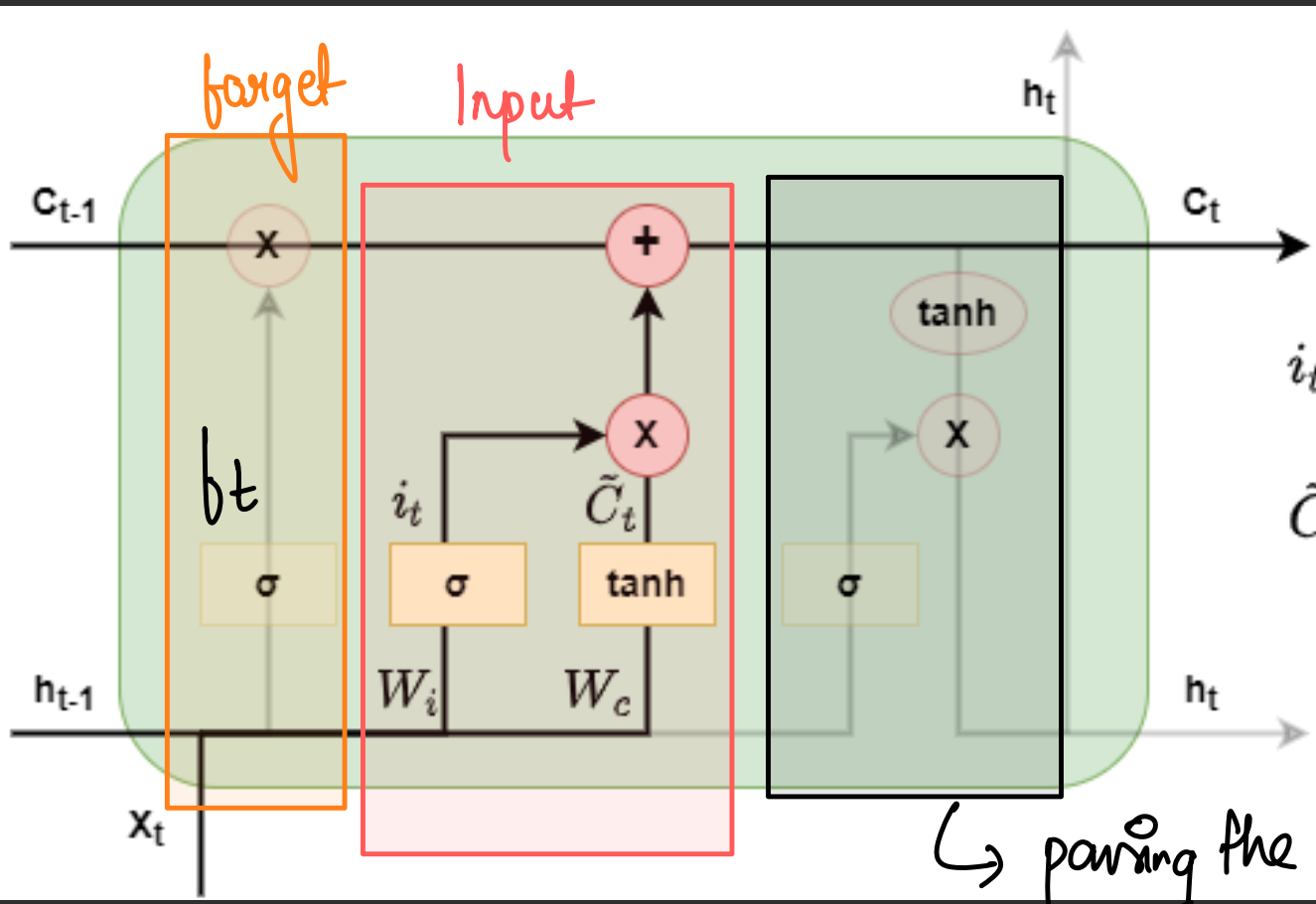
Matsix



Dense(64) → 64 Node

TimeDistributed(Dense(64))





$$i_t = \sigma(W_i * [h_{t-1}, x_t] + b_i)$$

$$\tilde{C}_t = tanh(W_C * [h_{t-1}, x_t] + b_c)$$

↳ passing the short term info to next

$$C_t = f_t \times C_{t-1} + i_t \times \tilde{C}_t$$

forget　　Input

{ forget some old info
add some new info

Eg: The movie was not good but the ending was amazing.

$$C_t = f_t \cdot C_{t-1} + i_t * \overline{C}_t$$

forget — Input

S_1: The X — $C_1 = [\ ]$

S_2: movie — C_2: [movie]

S_3: was X — C_3: [movie]

S_4: not — C_4: [movie, not]

S_5: good

C_5: [movie, not, ~~weak point~~ good]

not good

S_6: but

C_6: [movie, not, but]